

CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 16: Understanding Shapes

Topic-wise short notes with solved examples

Curves and polygons, convex and concave polygons, regular polygons, the sum of the interior and exterior angles of a polygon, and the number of diagonals.

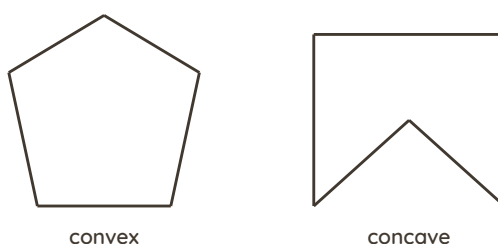
## 1 Curves and polygons

- A **simple curve** does not cross itself. A **closed curve** begins and ends at the same point.
- A **polygon** is a simple closed curve made only of line segments. The segments are its **sides** and their end points its **vertices**.
- A **diagonal** joins two vertices that are not next to each other.
- **Adjacent sides** share a vertex. **Adjacent vertices** are the ends of one side.

| Number of sides | 3        | 4             | 5        | 6       | 7        | 8       | 9       | 10      |
|-----------------|----------|---------------|----------|---------|----------|---------|---------|---------|
| Name            | Triangle | Quadrilateral | Pentagon | Hexagon | Heptagon | Octagon | Nonagon | Decagon |

## 2 Convex and concave polygons

- **Convex polygon**: every interior angle is less than  $180^\circ$ . All its diagonals lie inside it.
- **Concave polygon**: at least one interior angle is greater than  $180^\circ$  (a reflex angle). At least one diagonal lies partly outside.
- Unless told otherwise, "polygon" means a convex polygon.



A convex pentagon and a concave pentagon

## 3 Regular polygons

- A **regular polygon** has **all sides equal** and **all angles equal**. Examples: equilateral triangle, square, regular hexagon.
- A rhombus has equal sides but is not regular (its angles differ). A rectangle has equal angles but is not regular (its sides differ).

## 4 Interior angles

- A polygon of  $n$  sides can be divided into  $n - 2$  triangles by the diagonals from one vertex.

Sum of the interior angles =  $(n - 2) \times 180^\circ = (2n - 4)$  right angles

| Polygon       | Sides | Sum of interior angles | Each angle if regular |
|---------------|-------|------------------------|-----------------------|
| Triangle      | 3     | $180^\circ$            | $60^\circ$            |
| Quadrilateral | 4     | $360^\circ$            | $90^\circ$            |
| Pentagon      | 5     | $540^\circ$            | $108^\circ$           |
| Hexagon       | 6     | $720^\circ$            | $120^\circ$           |
| Octagon       | 8     | $1080^\circ$           | $135^\circ$           |
| Decagon       | 10    | $1440^\circ$           | $144^\circ$           |

- Each interior angle of a regular polygon =  $\frac{(n - 2) \times 180^\circ}{n}$ .

**Example 4.1** Find the sum of the interior angles of a polygon with 12 sides.

SOLUTION

$$(12 - 2) \times 180^\circ$$

$$1800^\circ$$

**Example 4.2** The sum of the interior angles of a polygon is  $1260^\circ$ . How many sides does it have?

SOLUTION

$$(n - 2) \times 180 = 1260, \text{ so } n - 2 = 7$$

$$9 \text{ sides}$$

**Example 4.3** Four angles of a pentagon are  $100^\circ$ ,  $110^\circ$ ,  $95^\circ$  and  $125^\circ$ . Find the fifth.

SOLUTION

$$540^\circ - 430^\circ$$

$$110^\circ$$

**Example 4.4** The angles of a hexagon are  $x$ ,  $x + 10^\circ$ ,  $x + 20^\circ$ ,  $x + 30^\circ$ ,  $x + 40^\circ$  and  $x + 50^\circ$ . Find  $x$ .

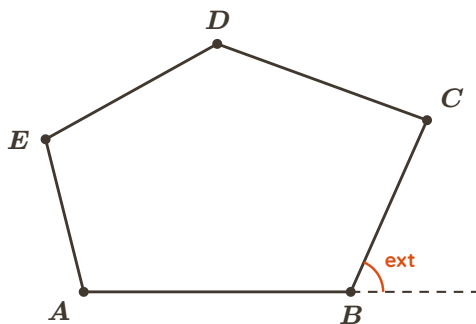
SOLUTION

$$6x + 150 = 720$$

$$x = 95^\circ$$

## 5 Exterior angles

- An **exterior angle** is formed when a side is produced. At each vertex, interior angle + exterior angle =  $180^\circ$ .
- The sum of the exterior angles of any convex polygon, one at each vertex, is  $360^\circ$ .
- For a regular polygon of  $n$  sides: each exterior angle =  $\frac{360^\circ}{n}$ , so  $n = \frac{360^\circ}{\text{exterior angle}}$ .
- Each interior angle of a regular polygon =  $180^\circ - \frac{360^\circ}{n}$ .



The exterior angle at B is formed by producing AB

**Example 5.1** Find each exterior angle and each interior angle of a regular polygon of 15 sides.

SOLUTION

Exterior:  $360^\circ \div 15$ . Interior:  $180^\circ - 24^\circ$ .

**$24^\circ$  and  $156^\circ$**

**Example 5.2** Each exterior angle of a regular polygon is  $40^\circ$ . How many sides does it have?

SOLUTION

$360 \div 40$

**9**

**Example 5.3** Each interior angle of a regular polygon is  $140^\circ$ . Find the number of sides.

SOLUTION

Exterior angle =  $40^\circ$

**9**

**Example 5.4** Can a regular polygon have each exterior angle equal to  $50^\circ$ ?

SOLUTION

$360 \div 50 = 7.2$ , which is not a whole number.

**No**

**Example 5.5** The ratio of each interior angle to each exterior angle of a regular polygon is 4 : 1. Find the number of sides.

SOLUTION

$4x + x = 180^\circ$ , so the exterior angle is  $36^\circ$ .

**10 sides**

## 6 Diagonals

- From one vertex of an  $n$ -sided polygon,  $n - 3$  diagonals can be drawn.

$$\text{Number of diagonals} = \frac{n(n-3)}{2}$$

**Example 6.1** How many diagonals does an octagon have?

SOLUTION

$$\frac{8 \times 5}{2}$$

**20**

**Example 6.2** A polygon has 35 diagonals. How many sides does it have?

SOLUTION

$$n(n-3) = 70 = 10 \times 7$$

**10 sides**

## 7 Common mistakes

- Using  $(n-2) \times 180^\circ$  for the sum of the exterior angles. That sum is always  $360^\circ$ .
- Taking  $n \times 180^\circ$  as the interior angle sum.
- Calling a rhombus or a rectangle a regular polygon.
- Finding the number of sides from the interior angle by dividing  $360^\circ$  by it. Use the exterior angle.
- Counting sides as diagonals.
- Accepting a number of sides that is not a whole number.